

Mehak Shaikh Mansoori

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Education

Indira Gandhi Delhi Technical University for Women

2023-2027

Bachelor of Technology - Electronics and Communication Engineering with Artificial Intelligence

CGPA: 9.23

Technical Skills

Languages: Java, Python, C, HTML, CSS

Tools & Technologies: Git, GitHub, Hugging Face, Transformers, Jupyter Notebook, Google Colab, VS Code, PyCharm

AI & ML: Supervised & Unsupervised Learning, Prompt Engineering, Generative AI, Model Evaluation, Random Forest

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating System

Databases: MySQL

Experience

Summer Intern

May 2026 – July 2026

Indian Institute of Technology, Kharagpur (IIT KGP)

- Conducted research on **Large Language Models (LLMs)**, focusing on **resource-aware benchmarking**, **prompt engineering**, and **LLM inference performance analysis**.
- Built a **benchmark dataset** of **1,000+ real-world prompts** spanning diverse **domains**, **reasoning strategies**, and **complexity levels** to profile **LLM inference workloads**.
- Profiled and compared **Mistral-7B** and **Qwen** by measuring **latency**, **memory usage**, **throughput**, and **token-level metrics** to analyze **resource utilization**.

Research Intern

June 2025 - July 2025

Defence Research and Development Organisation (DRDO) - ISSA LAB

- Optimized **real-time path planning** for **dynamic navigation scenarios**, improving **system efficiency** by **10%** through performance optimization and enabling **fast, data-driven decision-making**.

Projects

LeetTrackAI - [GitHub](#)

- Built a **full-stack web application** entirely using **prompt engineering** (no manual coding) leveraging **AI tools (Codex)**.
- Deployed the application for real-world usage and conducted testing to ensure **reliability and performance**.
- Tech Stack:** Prompt Engineering, OpenAI Codex, AI-Assisted Development

Robot Path Planning- [GitHub](#)

- Implemented a **scalable replanning framework** for **grid-based navigation**, supporting **large environments (1000×1000)**.
- Leveraged core **data structures and algorithmic** optimizations to ensure **performance and reliability**.
- Tech Stack:** Java, D Star Lite Algorithm, Data Structures and Algorithms.

PrefiSense- [GitHub](#)

- Developed a **machine learning pipeline** to predict **human preferences** between **chatbot responses**, achieving **81.94% accuracy** using a **Random Forest** classifier.
- Built a user-friendly **Streamlit application** that enables users to compare **chatbot responses** and obtain **real-time predictions**.
- Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Streamlit, Random Forest

Awards & Achievements

Solved **750+ DSA problems** on LeetCode and GeeksforGeeks, covering **core and advanced algorithms**.

Google India – The Big Code Challenge : Advanced to Round 2, ranking among the **Top 1,500 participants** 2026

Co-authored and Presented Research Paper: “ChatBot Arena Human Preferences Prediction” *ICICC 2025*

Selected for Amazon Future Engineer (AFE) Bootcamp 2025

Ernst and Young Foundation Bright Scholarship Recipient: Awarded to top 50 students in Delhi 2025

Winner – JPMC Generation Tech Hackathon: Chosen among 100 participants 2024

Other Relevant Experience

Student Mentor, Desh ke Mentor: Mentored 2 high school students, providing academic and career guidance.

Finance Intern, Slam Out Loud: Managed expense bills, ensuring timely ERP uploads & accurate record maintenance.